

③ Average Filter :-

Apply average/mean filter to the pixel indicated by (*) using a 3X3 neighborhood (matrix)

3	9	11	2
7	15*	8	8
10	12	9	10
1	9	11	2

$$\text{New Pixel} = (3 + 9 + 11 + 7 + 15 + 8 + 10 + 12 + 9) / 9$$
$$= 9$$

④ Median Filter :-

Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using 3X3 neighborhood using Median Filter.

1	8	8	0	7
4	7	9	5	7
5	4	⑥	8	6
4	2	0	1	5
0	1	0	2	0

We will arrange the values in ascending order.

0, 1, 2, 4, 5, 6, 7, 8, 9

The center element is 5 i.e. Median = 5.

So, 6 will be replaced by 5.